ASSIGNMENT 3: DRONE SOFTWARE SETUP & FLIGHT CONTROLLER CONFIGURATION

Full Name: Birendra Pratap Gautam

College Name: Rajkiya Engineering College Banda

Branch: Electrical Engineering

Year of Study: 3rd

Mobile Number: 8853746127

Email: birendrapratapgautam7@gmail.com

Internship Batch: B0826

submission Date: 01/09/2026

Q1. Flight Controller Configuration Analysis

A student assembled a quadcopter and installed the ArduPilot firmware using Mission Planner. During the first flight, the drone continuously tilted after takeoff, showed an incorrect heading, and did not respond properly to the radio transmitter. These problems are generally caused by incorrect configuration or calibration of the flight controller.

1. Accelerometer Calibration Error: If the accelerometer is not calibrated correctly, the flight controller cannot detect the drone's level position accurately. As a result, the drone tilts or drifts immediately after takeoff and may even crash. To fix this, connect the flight controller to Mission Planner, open Setup → Mandatory Hardware → Accelerometer Calibration, and complete the calibration by placing the drone in all the required positions.

2. Compass Calibration Error: An incorrectly calibrated compass causes the drone to display the wrong heading direction. This affects navigation and GPS-based flight modes. Perform Compass Calibration in Mission Planner away from metal objects.

3. Radio (RC) Calibration Error: If the radio transmitter is not calibrated, the drone will not respond correctly to throttle, pitch, roll, or yaw inputs. Calibrate all channels in

Mission Planner and verify the range is approximately 1000–2000 μs.

4. Flight Controller Orientation or Motor Configuration Error: Incorrect controller orientation or motor order can make the drone tilt or flip after takeoff. Verify controller orientation, motor numbering, motor rotation, and propeller installation.

Conclusion: Correct accelerometer, compass, radio calibration, and motor configuration ensure stable flight, accurate heading, and proper transmitter response.

Q2. Autonomous Flight & Safety Configuration

Feature	Role in Safe Autonomous Flight
Flight Modes	Auto, Loiter, Stabilize, and RTL control drone behaviour during different stages of the mission.
Return-To-Launch (RTL)	Automatically returns the drone to the take-off point during failsafe events or after mission completion.
Battery Failsafe	Monitors battery level and performs RTL or safe landing when voltage becomes low.
Radio Failsafe	Detects loss of transmitter signal and triggers RTL, Land, or another safe action.
GPS & Telemetry Monitoring	Provides real-time position, altitude, battery status, and mission monitoring.
Mission Planning	Defines waypoints, altitude, speed, and mission path for safe autonomous operation.

Effect of Losing Radio Signal and Incorrect Battery Failsafe

If the radio signal is lost and Radio Failsafe is not configured correctly, the drone may fly uncontrollably, drift away, or crash. If Battery Failsafe is disabled or configured incorrectly, the drone may continue flying until the battery is exhausted, causing an emergency landing or crash before returning home.

Conclusion

Proper configuration of Flight Modes, RTL, Battery Failsafe, Radio Failsafe, GPS & Telemetry Monitoring, and Mission Planning ensures a safe and reliable autonomous mission.

Ground Control Station (GCS) Software – Mission Planner

Mission Planner is one of the most widely used Ground Control Station (GCS) software applications for drones based on the ArduPilot platform. It is a free, open-source software that allows users to configure, monitor, control, and plan autonomous missions for unmanned aerial vehicles (UAVs). It provides an easy-to-use interface for both beginners and professionals working with drones.

1. Developer / Organization

Mission Planner was developed and is maintained by **Michael Oborne** as the primary Ground Control Station (GCS) for the **ArduPilot** open-source project. It is continuously improved with support from the global ArduPilot developer community.

2. Major Features

- **Live Telemetry:** Displays real-time flight information such as altitude, speed, GPS location, battery status, and flight mode.
- **Mission Planning:** Allows users to create and edit waypoints on a map for autonomous flights.
- **Parameter & PID Tuning:** Enables configuration and optimization of flight controller settings for better performance.
- **Firmware Installation:** Supports easy installation and updating of ArduPilot firmware.
- **Sensor Calibration Wizards:** Guides users through calibrating the compass, accelerometer, radio transmitter, and other sensors.
- **HUD (Heads-Up Display):** Provides a graphical display of the drone's attitude, heading, speed, altitude, and other flight data.
- **Flight Log Analysis:** Records and analyzes flight data to help diagnose issues and improve performance.

3. Supported Flight Controllers

Mission Planner supports a wide range of **ArduPilot-compatible flight controllers**, including:

- Pixhawk series
- Pixhawk-derivative boards

- CubePilot (Cube Orange, Cube Black)
- Other ArduPilot-compatible flight controllers

4. Key Advantages

- Completely **free and open-source**.
- Large and active **ArduPilot community** for support.
- Deep integration with ArduPilot firmware.
- Built-in calibration, configuration, and tuning tools.
- Reliable mission planning with real-time monitoring.
- Suitable for beginners, researchers, and professional drone operators.

5. Typical Applications

Mission Planner is widely used in:

- Agricultural drone operations and crop monitoring.
- Land surveying and mapping.
- Autonomous waypoint missions.
- Drone research and development (R&D).
- Educational and training projects.
- Inspection, aerial photography, and environmental monitoring.

Conclusion

Mission Planner is a powerful and user-friendly Ground Control Station software for ArduPilot-based drones. Its comprehensive features, support for multiple flight controllers, and open-source nature make it one of the best choices for drone configuration, mission planning, and flight monitoring. It is widely used in education, research, agriculture, surveying, and commercial UAV applications.